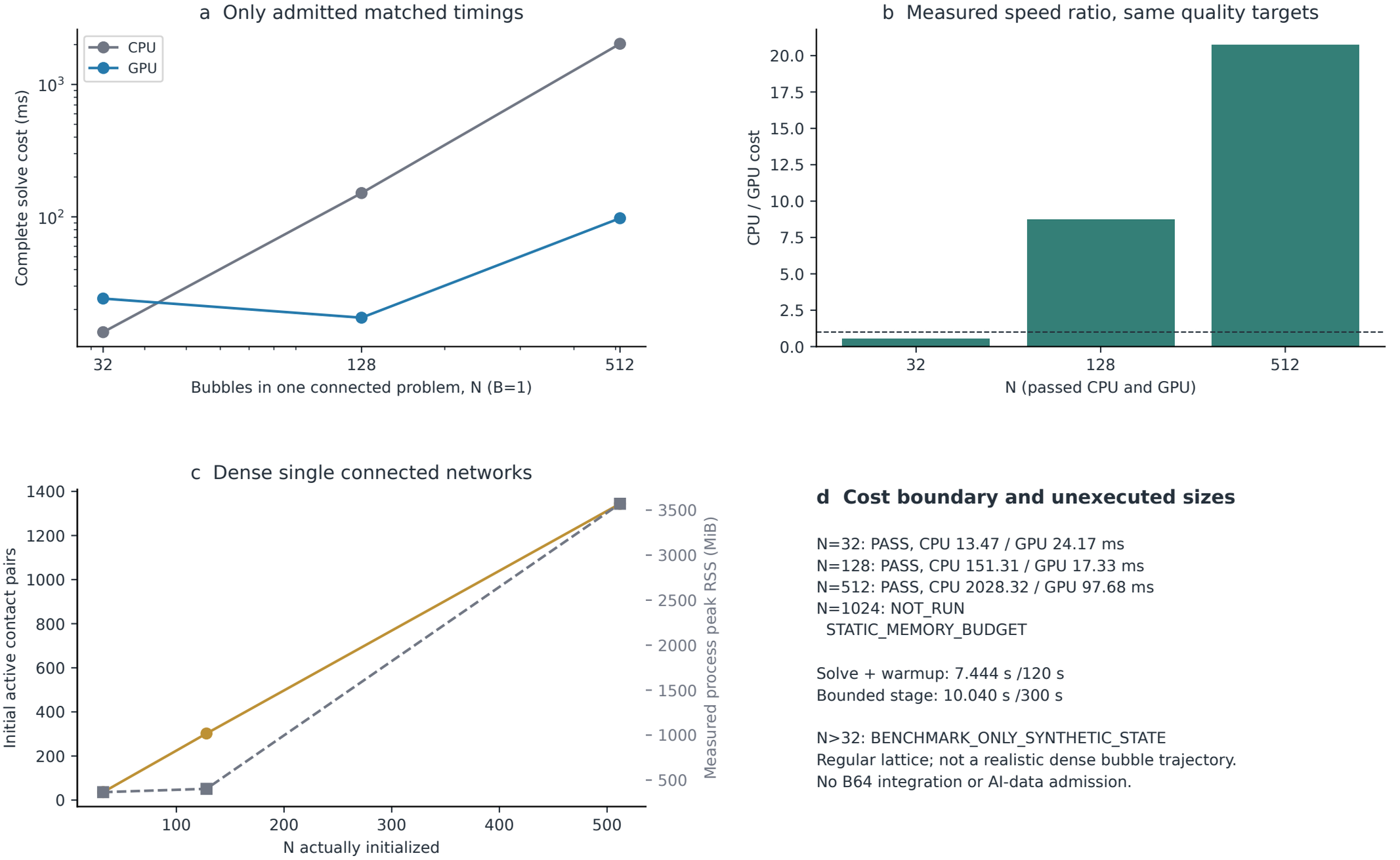
B2 fixed-state scaling: measured costs, numerical gates and explicit limits



d Cost boundary and unexecuted sizes

N=32: PASS, CPU 13.47 / GPU 24.17 ms
N=128: PASS, CPU 151.31 / GPU 17.33 ms
N=512: PASS, CPU 2028.32 / GPU 97.68 ms
N=1024: NOT_RUN
STATIC_MEMORY_BUDGET

Solve + warmup: 7.444 s /120 s
Bounded stage: 10.040 s /300 s

N>32: BENCHMARK_ONLY_SYNTHETIC_STATE
Regular lattice; not a realistic dense bubble trajectory.
No B64 integration or AI-data admission.

N=32 is real; 128/512 are regular synthetic networks with different iteration counts. BLAS 1 thread. Full timing includes preparation, residual checks and matched output.
One warmup + two formal solves; construction separate. Missing sizes are not measurements. Fixed-state numerical scaling is not physical validation.